

LANCASTER, TX, USA

WMO: 722143

Lat: 32.579N

Lon: 96.719W

Elev: 153

StdP: 99.50

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-2.6	-12.1	1.3	-1.0	-9.5	1.7	1.7	11.9	8.6	10.9	9.3	3.0	0	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	37.9	23.2	37.0	23.2	35.8	23.3	25.7	31.7	25.3	31.8	24.9	31.4	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.5	27.6	23.7	18.9	27.4	23.0	18.1	27.1	80.1	31.5	77.9	32.0	76.0	31.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.5	7.6	-8.6	39.8	2.2	1.7	-10.2	40.9	-11.5	41.9	-12.7	42.8	-14.4	44.0		
			-9.3	26.4	2.1	0.8	-10.8	27.0	-12.1	27.4	-13.3	27.9	-14.8	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	7.6	9.7	14.1	18.1	22.6	27.2	28.7	29.0	25.6	19.4	13.2	8.7	(6)
		DBStd	8.74	5.28	5.80	5.13	4.22	3.54	2.35	2.26	2.59	3.17	4.62	5.12	5.13	
		HDD10.0	318	109	71	21	2	0	0	0	0	2	25	88		
		HDD18.3	1300	333	246	146	55	9	0	0	1	43	166	302		
		CDD10.0	3494	35	63	149	243	390	516	580	468	293	122	47		
		CDD18.3	1434	1	4	16	47	141	266	322	330	218	76	13	2	
		CDH23.3	16176	11	25	98	353	1267	3008	4012	4198	2373	736	80	14	
		CDH26.7	8044	1	2	9	73	451	1504	2208	2372	1159	252	12	1	
(14)	Wind	WSAvg	3.0	3.3	3.6	3.8	3.8	3.2	2.9	2.4	2.3	2.2	2.7	3.1	3.1	(14)
(15)	Precipitation	PrecAvg	975	69	67	90	93	123	92	51	56	71	109	79	75	(15)
(16)		PrecMax	1683	159	187	179	185	386	273	125	129	237	298	258	221	(16)
(17)		PrecMin	530	5	6	2	30	15	17	0	0	12	7	12	6	(17)
(18)		PrecStd	254	41	45	48	46	71	66	37	35	55	77	59	50	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	25.0	26.2	27.8	31.2	34.0	37.4	39.1	40.1	37.3	33.3	28.0	25.2		(19)
(20)		MCWB	16.3	16.6	18.1	21.4	22.8	22.7	23.3	23.1	22.5	21.8	19.5	17.5		(20)
(21)		DB	22.1	23.6	26.0	28.4	32.4	36.1	37.6	38.3	35.7	31.5	25.6	22.5		(21)
(22)		MCWB	14.3	16.6	18.3	19.9	22.5	22.9	23.3	23.3	22.9	21.3	18.3	16.6		(22)
(23)		DB	19.0	21.5	24.1	27.1	31.1	34.8	36.5	37.4	34.1	29.6	23.3	20.0		(23)
(24)		MCWB	13.8	15.9	17.2	19.4	22.2	23.4	23.4	23.2	22.9	20.4	17.5	15.5		(24)
(25)		DB	17.2	19.7	22.4	25.3	29.2	33.3	35.2	36.0	32.7	27.5	21.9	17.7		(25)
(26)		MCWB	12.9	15.4	16.8	18.7	21.6	23.4	23.7	23.4	22.3	19.3	17.2	13.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	18.8	20.1	20.7	23.0	25.0	26.6	26.5	25.8	25.7	24.4	21.4	19.9		(27)
(28)		MCDB	22.0	23.5	24.4	28.0	30.3	31.6	31.7	32.1	30.9	28.4	24.8	23.1		(28)
(29)		WB	17.1	18.9	19.7	21.9	24.1	25.6	25.7	25.3	24.9	23.3	20.3	18.4		(29)
(30)		MCDB	19.0	21.7	23.5	26.3	29.5	31.2	31.8	32.8	30.8	28.2	23.2	21.2		(30)
(31)		WB	15.4	17.6	19.0	21.1	23.4	25.0	25.2	25.0	24.2	22.4	19.1	16.7		(31)
(32)		MCDB	18.3	20.1	22.6	24.8	28.7	31.0	31.8	32.4	30.2	27.0	22.4	19.0		(32)
(33)		WB	12.9	15.9	18.0	20.3	22.8	24.5	24.8	24.5	23.7	21.5	17.8	14.6		(33)
(34)		MCDB	16.6	18.9	21.3	24.0	27.7	30.4	31.3	31.9	29.5	25.5	21.1	17.3		(34)
(35)	Mean Daily Temperature Range	MDBR	11.4	11.2	11.6	11.8	11.1	11.2	11.8	12.1	12.2	12.6	11.8	11.1		(35)
(36)		5% DB	15.0	14.2	12.8	12.6	12.6	12.9	13.3	13.9	13.6	14.0	13.2	14.4		(36)
(37)		MCWBR	9.7	8.9	7.2	6.3	4.6	3.6	3.3	3.5	4.3	5.8	7.7	9.6		(37)
(38)		5% WB	12.5	11.4	10.6	10.1	10.4	10.5	11.3	11.8	11.3	11.1	10.9	11.8		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.317	0.332	0.359	0.399	0.428	0.441	0.444	0.445	0.403	0.360	0.341	0.321		(40)
(41)		taud	2.499	2.462	2.405	2.314	2.291	2.288	2.319	2.297	2.368	2.473	2.468	2.489		(41)
(42)		Ebn at Noon	905	923	919	891	863	848	844	838	863	881	867	878		(42)
(43)		Edh at Noon	86	99	113	129	133	133	129	130	116	97	89	83		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.86	3.31	4.42	5.38	5.80	6.53	6.67	6.13	5.16	4.06	3.04	2.53		(44)
(45)		RadStd	0.28	0.53	0.36	0.40	0.50	0.61	0.46	0.41	0.43	0.60	0.38	0.33		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page